

19 Some Properties of RG Flow

In the previous Section we have discussed the basic properties of the Gaussian fixed point $\mathcal{A}_{*G}$ at different dimensions d . At special values of d the fixed point theory has marginal operators. Thus, at $d = 4$ the field $\Phi_4 \sim \varphi^4$ is marginal, and as more careful calculation shows, it is *marginally irrelevant*. This is expressed by the RG flow equation (18.16) (note that it is equivalent to the leading term in the beta function (15.19)).

RG flow and Poincaré-Dulac normal form

It is interesting to see, in general, what is special about the marginal deformations from the mathematical point of view. Discussing on the abstract level, let us assume that we have generic critical fixed point $\mathcal{A}_* \in \Sigma$. Let y^α be some local coordinates on Σ , covering some neighborhood of $\mathcal{A}_*$ and chosen in such a way that the fixed point corresponds to $y^\alpha = 0$. One can think of y^α as the collection of coupling constants $u_{2k} - u_{2k,*}, v_{2k} - v_{2k,*}, \dots$, where u 's, v 's, etc, are the coefficients appearing in writing $\mathcal{A}$ in terms of various monomials in φ_0 and its derivatives, and u_* 's, etc, are their fixed-point values. Anyhow, the RG flow equation

$$\frac{d}{dl}\mathcal{A} = B\{\mathcal{A}\}$$

can be written in such coordinates as

$$\frac{d}{dl}y^\alpha = B^\alpha(\{y\}), \quad (19.1)$$

where B^α are some functions of all y 's. As $y = 0$ is assumed to be a fixed point, these functions satisfy the condition

$$B^\alpha(\{0\}) = 0.$$

Let us assume that these functions can be expanded into Taylor series around the fixed point,

$$B^\alpha(\{y\}) = K_\beta^\alpha y^\beta + C_{\beta\gamma}^\alpha y^\beta y^\gamma + \dots,$$

where K , C , etc, are some constants, and summation over repeated indices is assumed. The coefficients K_β^α are the matrix elements of the operator K , the linearization of the RG generator B near the fixed point,

$$B\{\mathcal{A}_* + \delta\mathcal{A}\} = K \delta\mathcal{A} + O(\delta\mathcal{A}^2).$$

By appropriate linear transformation of the coordinates y : $y^\alpha \rightarrow M_\beta^\alpha y^\beta$ the matrix K_β^α can be brought to diagonal form¹⁹. In these coordinates the flow equation (19.1) takes the form

$$\frac{d}{dl}y^\alpha = \kappa_\alpha y^\alpha + c_{\beta\gamma}^\alpha y^\beta y^\gamma + \cdots, \quad (19.2)$$

where κ_α are the eigenvalues of K , and no summation over α is assumed in the first term in the r.h.s.; the coefficients c are related to C by a linear transformation, in an obvious way.

No further simplification of the linear term in the r.h.s. of (19.2) is possible. However one may ask if the quadratic and higher order terms in the r.h.s. can be simplified by nonlinear coordinate transformation

$$y^\alpha = y^\alpha(\{\tilde{y}\}) = \tilde{y}^\alpha + a_{\beta\gamma}^\alpha \tilde{y}^\beta \tilde{y}^\gamma + \cdots.$$

Under coordinate transformations the functions B^α transform as components of a vector field on Σ . Simple calculation shows that in terms of $\tilde{y}$ the RG flow equations take the form

$$\frac{d}{dl}\tilde{y}^\alpha = \kappa_\alpha \tilde{y}^\alpha + \tilde{c}_{\beta\gamma}^\alpha \tilde{y}^\beta \tilde{y}^\gamma + \cdots,$$

where

$$\tilde{c}_{\beta\gamma}^\alpha = c_{\beta\gamma}^\alpha - (\kappa_\alpha - \kappa_\beta - \kappa_\gamma) a_{\beta\gamma}^\alpha.$$

One can completely eliminate the quadratic term by taking

$$a_{\beta\gamma}^\alpha = \frac{c_{\beta\gamma}^\alpha}{\kappa_\alpha - \kappa_\beta - \kappa_\gamma};$$

with this choice

$$\frac{d}{dl}\tilde{y}^\alpha = \kappa_\alpha \tilde{y}^\alpha + O(\tilde{y}^3).$$

This is true provided the combinations of the eigenvalues appearing in the denominators do not vanish. However if some triplet of the eigenvalues satisfy the *resonance condition*

$$\kappa_\alpha - \kappa_\beta - \kappa_\gamma = 0$$

the “resonance” term

$$c_{\beta\gamma}^\alpha y^\beta y^\gamma$$

¹⁹Non-diagonalizable forms contradict unitarity (positivity of norms)

(no summation) cannot be eliminated.

Similar situation takes place in the higher orders. If components B^α of a vector field B can be written as power series

$$B^\alpha(\{y\}) = \kappa_\alpha y^\alpha + c_{\beta_1\beta_2}^\alpha y^{\beta_1} y^{\beta_2} + c_{\beta_1\beta_2\beta_3}^\alpha y^{\beta_1} y^{\beta_2} y^{\beta_3} + \dots \quad (19.3)$$

in some local coordinates y^α , then by means appropriate analytic coordinate transformation

$$y^\alpha \rightarrow y^\alpha + a_{\beta_1\beta_2}^\alpha y^{\beta_1} y^{\beta_2} + a_{\beta_1\beta_2\beta_3}^\alpha y^{\beta_1} y^{\beta_2} y^{\beta_3} + \dots \quad (19.4)$$

one can exterminate all nonlinear terms in (19.3), except for the so-called resonance terms. A term

$$c_{\beta_1\beta_2\dots\beta_n}^\alpha y^{\beta_1} y^{\beta_2} \dots y^{\beta_n} \quad (19.5)$$

in the function B^α is called the *resonance term* if the following *resonance condition*

$$\kappa_\alpha = \kappa_{\beta_1} + \kappa_{\beta_2} + \dots + \kappa_{\beta_n} \quad (19.6)$$

is satisfied. One can always adjust the coefficients a in (19.4), order by order in y , in such a way that in new coordinates

$$\frac{d}{dl} y^\alpha = \kappa_\alpha y^\alpha + \text{Resonance Terms} \quad (19.7)$$

If there are no resonances, all nonlinear terms in (19.3) can be exterminated. This statement is known as *Poincare-Dulac theorem*²⁰. I will call “canonical” the coordinates y in which the flow equation takes the form (19.7).

Two remarks are in order. First, the above theorem *does not* suggest that the global structure of the RG flow is in any sense “trivial”, even in non-resonance cases. The series (19.4) which defines the transformation converges at best in some finite neighborhood of the fixed point $y = 0$. Another interesting irregularities the global RG flow may possess (like another fixed points) are located no closer then at the boundary of this domain of convergence; in the new coordinates where B^α ’s are “trivial” they are “pushed out to the infinity”. To get some idea consider the following

²⁰See e.g. V.I Arnold, Geometrical Methods in the Theory of Ordinary Differential Equations, Springer, 1988

Exercise: Study the flow equation with just one coordinate,

$$\frac{d}{dl} y = y - y^2$$

(which has two fixed points $y = 0$ and $y = 1$). Find transformation $y = f(\tilde{y}) = \tilde{y} + \dots$ such that

$$\frac{d}{dl} \tilde{y} = \tilde{y}.$$

Study the properties of the map $y \rightarrow \tilde{y}$.

Second remark concerns very simple interpretation which the above resonance conditions have in terms of ordinary dimensional analysis. In the neighborhood of $\mathcal{A}_*$ where the coordinates y are defined, the action can be written as

$$\mathcal{A} = \mathcal{A}_* + y^\alpha \int d^d x \Phi_\alpha(x) + O(y^2),$$

where $\Phi_\alpha(x)$ are the fields which diagonalize the operator D associated with the fixed point $\mathcal{A}_*$. The eigenvalues D_α - i.e. the dimensions of the fields Φ_α - are related to κ_α as

$$D_\alpha + \kappa_\alpha = d.$$

Therefore the numbers κ_α can be interpreted as the mass dimensions of the “coupling constants” y^α . The resonance condition (19.6) then states that the resonance term (19.5) has “right dimension”. In this sense, the coordinate system in which the RG flow equations have the form (19.7) respects the straightforward dimensional analysis.

From this point of view, what distinguishes the marginal case is that the presence of a marginal field (i.e. the field Φ_α with $D_\alpha = d$, or $\kappa_\alpha = 0$) always leads to infinitely many resonances. Indeed, if, say, $\kappa_0 = 0$, the the resonance condition (19.6) with any α can be satisfied by taking $\beta_1 = \alpha$ and $\beta_2 = \beta_3 = \dots = \beta_n = 0$.

To illustrate this point let us consider an example with only two coupling constants y^0, y^1 , such that $\kappa_0 = 0$ and $\kappa_1 = 2$. I will use suggestive notations $y^0 = \lambda$ and $y^1 = M^2$. Then in the flow equation for λ all terms λ^n are resonant, so the coordinates exist in which this equation takes the form

$$\frac{d}{dl} \lambda = \sum_{n=2}^{\infty} b_n \lambda^n = -\beta(\lambda). \quad (19.8)$$

Similarly, in the equation for M^2 all terms of the form $M^2 \lambda^n$ are resonant, and in canonical coordinates one can write

$$\frac{d}{dl} M^2 = 2 M^2 - \sum_{n=1}^{\infty} \gamma_n M^2 \lambda^n = (2 - \gamma_{\varphi^2}(\lambda)) M^2. \quad (19.9)$$

Note that with this notations these equations have exactly the form of the coupling and the mass “flow” equations associated with the Callan-Symanzik equation of the $d = 4$ φ^4 theory.

Remark: In fact, in this important example a freedom still remains for further changes of coordinates $\lambda \rightarrow \tilde{\lambda}(\lambda)$, $M^2 \rightarrow \tilde{M}^2 = M^2 f(\lambda)$, which doesn’t change the above “canonical” form of the flow equations.

Exercise: Check (to the order $n =$ when you get bored, or see the pattern) that all terms $\sim \lambda^n$ with $n \geq 4$ can be exterminated order by order, by suitable choice of coefficients a_n in

$$\tilde{\lambda} = \lambda + a_3 \lambda^3 + a_4 \lambda^4 + \dots.$$

Further, check that suitable redefinition of $M^2 \rightarrow \tilde{M}^2$,

$$\tilde{M}^2 = M^2 (1 + c_1 \lambda + c_2 \lambda^2 + \dots)$$

allows one to eliminate the terms with $n \geq 2$ in the r.h.s. of (19.9). In reference to renormalization of the φ^4 theory at $d = 4$, these results imply that a scheme exists in which all n th order corrections to the beta-function vanish starting from $n = 4$, and all corrections to γ_{φ^2} vanish starting from $n = 2$. At the same time the coefficients b_2 and b_3 , and γ_1 , are scheme-independent.

Note that the statement of the Poincare-Dulac theorem concerns with formal power series. Existence of “canonical coordinates”, even locally, depends on convergence of the series which in general is not guaranteed. For general RG flow in the full space Σ the situation is indeed difficult. One of the problems lays again in the “resonance denominators” $\kappa_\alpha - \kappa_{\beta_1} - \dots - \kappa_{\beta_n}$ which the coefficients a in the series (19.4) have. When both positive and negative κ_α are present, even if no resonance conditions (19.6) are satisfied exactly, they often are “almost satisfied”, in which case some denominators come very close to zero giving rise to anomalously large coefficients in (19.4).

The situation becomes much more favorable if, instead of the flow in the full space Σ , we concentrate attention on the RG flow in an unstable manifold $U(\mathcal{A}_*)$

associated with some fixed point $\mathcal{A}_*$. First, $U(\mathcal{A}_*)$ typically has finite dimensionality (as our study of $\mathcal{A}_{*,G}$ illustrates). Also, by the definition of unstable manifold all corresponding κ are either positive (relevant) or zero (marginal). In this case the resonance denominators are under much better control and the series (19.4) defining the canonical coordinates usually converges in finite neighborhood of $\mathcal{A}_*$. Recall that $U(\mathcal{A}_*)$ is a part of $\Sigma(\infty)$, i.e. it describes finite local QFT. For local QFT generated by the fixed point $\mathcal{A}_*$ the canonical coordinates do exist.

Callan-Symanzik equation revisited

The preceding analysis allows one to re-derive the Callan-Symanzik equation which we have obtained before on the basis of renormalized perturbation theory.

Let $\mathcal{A}_*$ be a fixed point. Assuming that its unstable manifold is n -dimensional, let $\{\lambda^i\}, i = 1, \dots, n$ be coordinates on $U(\mathcal{A}_*)$, such that $\mathcal{A}_*$ corresponds to $\lambda = 0$. This unstable manifold describes n -parameter family of finite local field theories. I will denote the corresponding correlation functions as

$$\langle \varphi(x_1) \cdots \varphi(x_N) \rangle_\lambda,$$

indicating their dependence on n parameters λ^i .

In the coordinates λ the RG flow equation can be written as

$$\frac{d}{dl} \lambda^i = -\beta^i(\lambda), \quad (19.10)$$

where $-\beta^i$ are components of the vector field B^α on $U(\mathcal{A}_*)$ (by definition of U , $B(U)$ is tangent to U), the minus sign is taken to match our previous notation for the beta-function. According to the definition of the Wilson's RG, if $\lambda(l)$ is a solution of (19.10) with some initial condition $\lambda(0) = \lambda$, the field theories corresponding to λ and $\lambda(l)$ are related by the scale transformation $x \rightarrow x/L$. In particular, one can write for the correlation functions

$$\langle \varphi(x_1) \cdots \varphi(x_N) \rangle_{\lambda(l)} = Z^{\frac{N}{2}}(L) \langle \varphi(Lx_1) \cdots \varphi(Lx_N) \rangle_{\lambda(0)},$$

where $L = e^l$. Taking infinitesimal transformation $l = \delta l$, $L = 1 + \delta l$, $\lambda(l) = \lambda - \beta(\lambda) \delta l$, we have

$$\left[x \frac{\partial}{\partial x} + N D(\lambda) + \sum_{i=1}^n \beta^i(\lambda) \frac{\partial}{\partial \lambda^i} \right] \langle \varphi(x_1) \cdots \varphi(x_N) \rangle_\lambda = 0, \quad (19.11)$$

where $x \frac{\partial}{\partial x}$ is the shorthand notation for

$$x \frac{\partial}{\partial x} = \sum_{k=1}^N x_k^\mu \frac{\partial}{\partial x_k^\mu},$$

and

$$D(\lambda) = L \frac{d}{dL} Z^{\frac{1}{2}}(L)|_{L=1}.$$

Note that $D(0) = D_\varphi$, the dimension of φ at the fixed point $\mathcal{A}_*$. $D(\lambda)$ can be written as

$$D(\lambda) = D_\varphi + \gamma(\lambda),$$

so that $\gamma(0) = 0$.

The equation (19.11) generalizes straightforwardly to correlation functions involving composite fields $\mathcal{O}_\alpha$. Repeating the same analysis we obtain

$$\begin{aligned} \sum_{k=1}^N \langle \mathcal{O}_{\alpha_1}(x_1) \cdots (\mathcal{DO})_{\alpha_k}(x_k) \cdots \mathcal{O}_{\alpha_N}(x_N) \rangle_\lambda + \\ \sum_{i=1}^n \beta^i(\lambda) \frac{\partial}{\partial \lambda^i} \langle \mathcal{O}_{\alpha_1}(x_1) \cdots \mathcal{O}_{\alpha_N}(x_N) \rangle_\lambda = 0, \end{aligned} \quad (19.12)$$

where

$$(\mathcal{DO})_\alpha(x) = x^\mu \frac{\partial}{\partial x^\mu} \mathcal{O}_\alpha(x) + D_\alpha^\beta(\lambda) \mathcal{O}_\beta(x),$$

and

$$D_\alpha^\beta(\lambda) = -L \frac{d}{dL} z_\alpha^\beta(L)|_{L=1};$$

$z_\alpha^\beta(L)$ describes the RG transformation of the composite fields $\mathcal{O}_\alpha$ (see Sect.18). Again,

$$D_\alpha^\beta = D_\alpha^\beta(\lambda)|_{\lambda=0}$$

coincides with the matrix of dimensions at $\mathcal{A}_*$, whose eigenvalues are the dimensions of the fields Φ_α .

We see that the CS equation is very general consequence of Wilson's RG theory and not just the property of the perturbation theory around the Gaussian fixed point (i.e. free field theory).

20 Fixed Points and Criticality

It is possible to establish direct relation between the singular behavior of thermodynamic quantities near critical points and the fixed points of the Wilson's RG. This relation explains experimentally observed *universality* of critical singularities.

Let us consider again a magnet. As we have seen few lectures ago, the long-range fluctuations of local magnetization $M(x)$ can be taken into account by considering the statistical integral

$$Z = \int [\mathcal{D}\varphi_0] e^{-\mathcal{A}_0[\varphi_0]},$$

where the field $\varphi_0(x)$ is proportional to $M(x)$,

$$\varphi_0(x) = C(T) M(x),$$

with some temperature-dependent coefficient. It is assumed that this functional integral has cutoff, with the cutoff parameter Λ such that

$$\Lambda^{-1} \gg a,$$

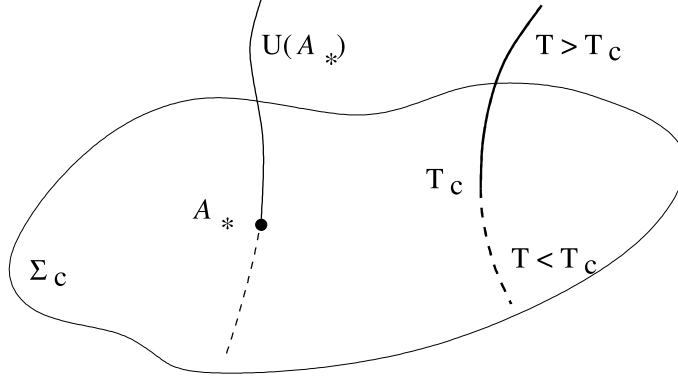
the atomic scale. Exact form of the action $\mathcal{A}_0$ is not known; it combines statistical contributions of all microscopic fluctuations and depend significantly on all details of the microscopic interactions. In Landau-Ginsburg approximation one takes

$$\mathcal{A}_0 = \int d^d x \left[\frac{1}{2} (\partial \varphi_0)^2 + \frac{m_0^2(T)}{2} \varphi_0^2 + \frac{\lambda_0(T)}{4!} \varphi_0^4 \right],$$

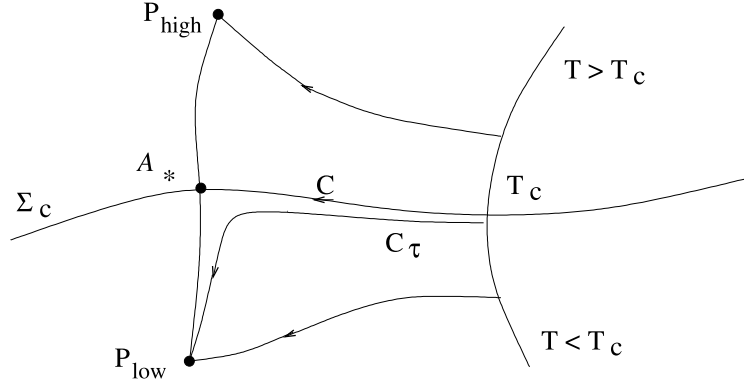
where $m_0^2(T)$, $\lambda_0(T)$ are unknown functions of T , but it can be any other combination of powers of φ_0 and its derivatives. What is important, all the coefficients are some (generally unknown) functions of the temperature T . This means that the actions describing real physical system, with all its microscopic details, being functions of the temperature, form one-dimensional submanifold in Σ ; I will call it the “physical curve” (Wilson used the term “canonical curve”).

Now let us assume that there is a fixed point $\mathcal{A}_*$ out there in Σ . At this point we actually will consider a subspace $\Sigma_{\text{symm}} \subset \Sigma$ which contains only actions invariant under $\varphi_0 \rightarrow -\varphi_0$. We assume that there is a fixed point $\mathcal{A}_* \in \Sigma_{\text{symm}}$ such that it has only one (symmetric) relevant (or marginally relevant) field Φ_0 , with dimension $D_0 < d$. In other words, the unstable manifold in Σ_{symm} is one-dimensional, and correspondingly the critical surface Σ_c has codimension one. This means that

the physical curve generally intersects this critical surface at some point. This is illustrated in the picture below.



Let us denote T_c the temperature corresponding to the intersection point. The large-distance properties of the system can be understood if we can follow the RG flow which starts at any point of the physical curve. Let us consider a typical pattern of the RG flow in this situation. Besides the critical fixed point $\mathcal{A}_*$, there are two non-critical ($R_c = 0$) fixed points, P_{high} and P_{low} ²¹.



A trajectory starting at any point on the physical curve corresponding to $T > T_c$, and far enough from T_c , converges quickly to the noncritical fixed point P_{high} . Similarly, starting at any point with $T < T_c$ one quickly reaches P_{low} . The only trajectory which behaves qualitatively differently is the one that starts at the point T_c itself. This trajectory C lays entirely at the critical surface, and converges to the

²¹For a magnetic transition P_{high} corresponds to the probability distribution $P_{\text{high}}[\varphi_0] \sim \prod_x \delta(\varphi_0(x))$, and $P_{\text{low}}[\varphi_0] \sim \prod_x \delta(\varphi_0(x) - \bar{\varphi}_0) + \prod_x \delta(\varphi_0(x) + \bar{\varphi}_0)$

critical fixed point $\mathcal{A}_*$ as $l \rightarrow \infty$. This means that the large-distance physics exactly at the critical temperature is completely determined by the fixed-point action $\mathcal{A}_*$. In particular, the two-point correlation function will have the scale-invariant form

$$\langle M(x)M(x') \rangle_{T=T_c} = \frac{\text{Const}}{|x - x'|^{2D_\varphi}}$$

for $|x - x'| \gg a e^{l_0}$, where l_0 is some finite amount of the RG “time” l needed to arrive at close proximity of $\mathcal{A}_*$. Here D_φ is the dimension of φ at this fixed point.

Consider now a trajectory C_τ which starts at some point T very close to T_c ,

$$\tau = \frac{T_c - T}{T_c} \ll 1. \quad (20.1)$$

To be definite I will assume that $T < T_c$. There are three distinct stages of the RG “evolution” for such trajectory. At the first stage it goes very close to C approaching $\mathcal{A}_*$, until it comes to close proximity of this fixed point. It then stays near $\mathcal{A}_*$ for very long RG “time” l . This is the second stage. Finally, in the third stage it departs toward P_{low} staying close to the unstable manifold U .

The second stage is the most important one; that is where the critical singularities are formed. Because at this stage the trajectory stays close to the fixed point, we can use linear analysis. Let us assume that in some finite amount l_0 of RG “time” the trajectory C_τ comes close to $\mathcal{A}_*$, where the corresponding action can be written as

$$\mathcal{A} = \mathcal{A}_* + y_0 \int d^d x \Phi_0(x) + \sum_{\alpha \neq 0} y_\alpha \int d^d x \Phi_\alpha(x), \quad (20.2)$$

where the constants y depend on τ , the deviation (20.1) from the critical point. It is important to note that this dependence is analytic because it comes from integrating the RG flow equations for finite “time” l_0 . As for $\tau = 0$ one must stay at the critical surface, which in the linear approximation corresponds to $y_0 = 0$, for small τ we can assume

$$y_0 = \text{Const} \times (T_c - T), \quad (20.3)$$

and neglect the τ -dependence of all y_α with $\alpha \neq 0$.

In the second stage the RG flow is described by the equations

$$\frac{d}{dl} y_0 = \kappa_0 y_0, \quad \frac{d}{dl} y_\alpha = \kappa_\alpha y_\alpha,$$

where the coefficients κ are related to the dimensions D_α of the fields Φ_α ,

$$\kappa_0 = d - D_0 > 0, \quad \kappa_\alpha = d - D_\alpha < 0 \quad \text{for } \alpha \neq 0.$$

During this second stage all irrelevant terms in (20.2) die out, while the coefficient y_0 increases as

$$y_0(l) = y_0 e^{\kappa_0(l-l_0)}. \quad (20.4)$$

Recall that if R_c is the correlation length corresponding to the initial action on the physical curve, the correlation length corresponding to the time l along the trajectory is

$$R_c(l) = R_c e^{-l}.$$

From this and (20.4) one can find²²

$$R_c \sim y_0^{-\nu}, \quad (20.5)$$

where

$$\nu = \frac{1}{\kappa_0} = \frac{1}{d - D_0},$$

and from the linear dependence (20.3)

$$R_c \sim |T_c - T|^{-\nu}.$$

Thus, the correlation length diverges near T_c , with the exponent determined by the dimension D_0 of the relevant field Φ_0 . This divergence is in full agreement with the idea that it is the large-scale fluctuations which are responsible for the critical behavior.

Similar way one can relate other critical exponents to the dimensions of various fields at the fixed point $\mathcal{A}_*$. For example, the spontaneous magnetization

$$\bar{M} \sim \langle \varphi_0 \rangle$$

(nonzero in the low-T phase) must satisfy the CS equation, which in this case reads

$$\left[D_\varphi - \kappa_0 y_0 \frac{\partial}{\partial y_0} \right] \bar{M} = 0,$$

²²Observe that within the linear approximation (20.2) the ratio $R_c(l)/y_0^{-\nu}(l)$ is a constant independent of l .

i.e.

$$\bar{M} \sim y_0^{\frac{D_\varphi}{\kappa_0}}. \quad (20.6)$$

Again, taking into account (20.3) we find (for $T < T_c$)

$$\bar{M} \sim (T_c - T)^\beta,$$

where

$$\beta = \frac{D_\varphi}{d - D_0}.$$

It is useful to note that the relations (20.5), (20.6) can be obtained by simple dimensional counting, if one assumes that the coupling constant y_0 has the dimension κ_0 and the field φ_0 has the dimension D_φ . This follows from the CS equation with the linear approximation for the RG functions β ; in fact it is valid beyond the linear approximation, as it follows from the existence of “canonical coordinates”.

As another example of this dimensional counting let us find some other critical exponents in terms of dimensions. If, at $T = T_c$, one adds an external field, i.e. the term

$$H \int d^d x \varphi_0(x),$$

the dimensional counting implies that H has dimension $d - D_\varphi$, and so the magnetization as the function of H must have the following singular behavior

$$\bar{M} \sim H^{\frac{1}{\delta}},$$

where

$$\delta = \frac{d - D_\varphi}{D_\varphi}.$$

Similar way one finds for the specific heat at $H = 0$

$$C \sim |T_c - T|^\alpha$$

with

$$\alpha = 2 - d\nu. \quad (20.7)$$

Note that all these exponents are expressed through just two numbers, the dimensions of the fields φ and Φ_0 . Therefore no matter what the values of this dimensions

are, the critical exponents must obey certain *scaling relations*, the relation (20.7) being one example. These relation are confirmed in experiment.

Note that the above analysis relies on the properties of the fixed point $\mathcal{A}_*$, as well as on general topological properties of RG. Very little indeed depends on the details of the original actions on the “physical curve”, and hence on the details of the microscopic structure of the magnet. One expects therefore that the most important characteristics of the critical singularities must not be different for phase transitions in different magnetic materials (with the same symmetry). One fixed point may represent the criticality in the whole class of physical systems. It is said it describes a *universality class* of critical behavior. In fact, systems of very different physical nature can belong to the same universality class. For example, the liquid-gas critical point is likely to represent the same universality class as the phase transition in axial ferromagnet.

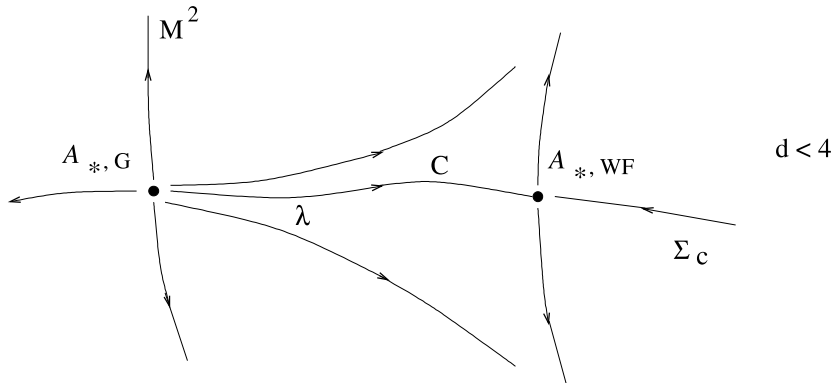
This all shows that understanding the fixed points is of crucial importance both in statistical mechanics and in quantum field theory. Unfortunately, analysis and classification of all fixed points are not yet achieved. In fact, at present very few nontrivial fixed points can be analyzed on a systematic level.

21 Wilson-Fisher Fixed Point

Previously, we have studied the simplest example of fixed point - the Gaussian fixed point corresponding to free massless scalar field. It is easy to argue, however, that the Gaussian fixed point cannot be responsible for the ordinary transition in a magnet in three dimensions (the Gaussian fixed point describes the so called “tricritical behavior”, as we will briefly discuss later). Indeed, it was very important for the arguments given in the previous lecture that the critical surface of $\mathcal{A}_*$ has codimension one in Σ_{symm} , i.e. that there is only one relevant field which respects the symmetry of the problem ($\varphi \rightarrow -\varphi$ in our case). However, as we have seen, the gaussian fixed point has more then one symmetric relevant fields for all $d < 4$. In particular, for $d = 3$ two fields, $\Phi_2 \sim \varphi^2$ and $\Phi_4 \sim \varphi^4$, are relevant. Therefore there must be another fixed point describing the universality class of the magnetic phase transition.

The best theoretical evidence for the existence of another nontrivial fixed point at $d = 3$ (on a formal level, at all $2 < d < 4$) was given by Wilson and Fisher, and it is usually referred to as the Wilson-Fisher fixed point $\mathcal{A}_{*WF}$.

At $d < 4$ the topology of the RG flow relating the fixed points $\mathcal{A}_{*G}$ and $\mathcal{A}_{*WF}$ is likely to be the following. The unstable manifold $U(\mathcal{A}_{*G})$ at $d < 4$ has dimensionality two; it is generated by the two relevant fields Φ_2 and Φ_4 (again, we restrict attention to the symmetric space Σ_{symm}). The fixed point $\mathcal{A}_{*WF}$ is likely to lay in $U(\mathcal{A}_{*G})$, so that the RG flow on this unstable manifold would look qualitatively as follows



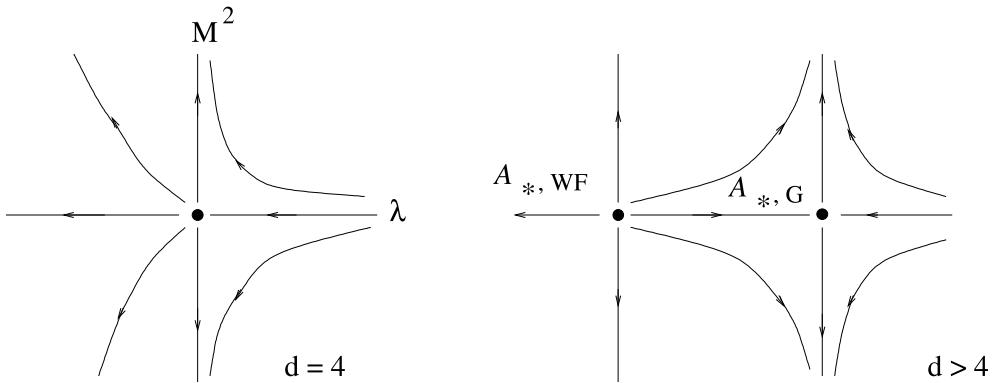
According to this picture there is a trajectory C connecting the fixed point $\mathcal{A}_{*G}$ to $\mathcal{A}_{*WF}$. Therefore $\mathcal{A}_{*G}$ lays on the critical surface Σ_c which also passes through $\mathcal{A}_{*WF}$. The unstable manifold $U(\mathcal{A}_{*WF})$ is then one-dimensional, as expected.

Note that the whole unstable manifold $U(\mathcal{A}_{*G})$ in this picture, being an element of $\Sigma(\infty)$, represent local field theories. Namely, it represents two-parameter family of φ^4 field theories. These field theories can be characterized by two parameters, the renormalized mass parameter M^2 and the renormalized coupling constant λ . Therefore these parameters can be taken for coordinates in the above picture. Although there is an ambiguity in the choice of these coordinates (the scheme dependence), the critical surface corresponding to $M^2 = 0$ is defined by the scheme-independent condition

$$\Gamma^{(2)}(p^2)|_{p^2=0} = 0. \quad (21.1)$$

In these coordinates the fixed point $\mathcal{A}_{*WF}$ lays at $M^2 = 0$ and $\lambda = \lambda_*$, where the value of λ_* depends on the scheme.

We can formally consider the φ^4 theory in d dimensional space, treating d as continuous parameter. What happens when d approaches 4? We know that at $d = 4$ the field φ^4 becomes marginally irrelevant, and for $d > 4$ it is strictly irrelevant, and so for all $d \geq 4$ the gaussian fixed point has only one relevant field φ^2 . This behavior emerges naturally if one assumes that when d approaches the value 4 from below the fixed point $\mathcal{A}_{*WF}$ gets very close to the gaussian fixed point, and at $d = 4$ these two fixed point merge. At $d > 4$ λ_* becomes negative, and it does not correspond to field theory in usual sense²³. This idea leads to the patterns of the RG flow at $d = 4$ and at $d > 4$ is shown in the pictures below.



Note that the pattern at $d = 4$ is exactly what one expects to see when φ^4 is marginally irrelevant, in which case we know that the flow in the vicinity of the

²³The “magnetic criticality” at $d \geq 4$ then is described by the gaussian fixed point.

gaussian fixed point is described by the equations

$$\begin{aligned}\frac{d}{dl} \lambda &= -\beta(\lambda) = -3 \frac{\lambda^2}{(4\pi)^2} + O(\lambda^3), \\ \frac{d}{dl} M^2 &= (2 - \gamma_{\varphi^2}(\lambda)) M^2 = \left(2 - \frac{\lambda}{(4\pi)^2} + O(\lambda^2)\right) M^2.\end{aligned}$$

On the basis of these arguments Wilson and Fisher have assumed that if one formally takes $d = 4 - \epsilon$ with ϵ small, say $d = 3.99$, the fixed point $\mathcal{A}_{*WF}$ can be observed within the framework of renormalized perturbation theory in φ^4 field theory.

Let us briefly discuss what changes in the renormalized perturbation theory of φ^4 if one takes $d = 4 - \epsilon$ instead of $d = 4$. As before, we start with the action in its “bare” form

$$\mathcal{A} = \int d^d x \left[\frac{1}{2} (\partial \varphi_0)^2 + \frac{m_0^2}{2} \varphi_0^2 + \frac{\lambda_0}{4!} \varphi_0^4 \right]$$

Note that in d dimensions the canonical dimension of φ_0 is $(d - 2)/2$ and hence coupling constant λ_0 has positive mass dimension,

$$[\varphi_0] = \frac{d - 2}{2} = 1 - \frac{\epsilon}{2}; \quad [\lambda_0] = 4 - d = \epsilon.$$

To develop renormalized perturbation theory we rewrite this action in terms of the renormalized field and renormalized parameters,

$$\begin{aligned}\mathcal{A} = \int d^d x \left[\frac{1}{2} (\partial \varphi)^2 + \frac{M^2}{2} \varphi^2 + \frac{\lambda}{4!} \varphi^4 + \right. \\ \left. \frac{\delta Z}{2} (\partial \varphi)^2 + \frac{\delta M^2}{2} \varphi^2 + \frac{\delta \lambda}{4!} \varphi^4 \right],\end{aligned}\tag{21.2}$$

where the counterterms are determined from suitably chosen normalization conditions.

It is worth stressing here that for $d < 4$ the diagrams describing the field and the coupling constant renormalizations do not diverge; the only divergent renormalization is the one of the mass parameter. Therefore the counterterms δZ and $\delta \lambda$ have finite limit when the cutoff parameter Λ goes to infinity. Therefore at $d < 4$

these renormalizations are introduced not for the purpose of exterminating the divergences, which are absent, but in order to introduce more suitable coordinate λ on $U(\mathcal{A}_{*G})$ instead of λ_0 . Although for $d < 4$ (unlike the case of $d = 4$) the bare coupling λ_0 can be used as local coordinate on this unstable manifold, we will see soon that it is not too useful for the purpose of studying the fixed point $\mathcal{A}_{*WF}$.

As the fixed point $\mathcal{A}_{*WF}$ is expected to be located on the critical surface, it suffices to study the massless φ^4 theory which is defined by the condition (21.1),

$$\Gamma^{(2)}(p^2)|_{p^2=0} = 0.$$

As before, choose some normalization scale μ and assume the following simple normalization conditions

$$\begin{aligned} \frac{d}{dp^2} \Gamma^{(2)}(p^2)|_{p^2=\mu^2} &= 1, \\ \Gamma^{(4)}(p_1, \dots, p_4)|_{[(p_i+p_j)^2=4\mu^2]} &= \lambda. \end{aligned} \quad (21.3)$$

The first of them fixes the normalization of φ while the second gives the definition of the renormalized coupling constant λ .

At $d < 4$ the Callan-Symanzik equation for the massless theory is

$$\left[x \frac{\partial}{\partial x} + N D(\lambda) + \beta(\lambda) \frac{\partial}{\partial \lambda} \right] \langle \varphi(x_1) \cdots \varphi(x_N) \rangle_\lambda = 0; \quad (21.4)$$

its form reflects the general properties of the RG we scratched the last time, which did not depend on d . However for generic d

$$D(\lambda) = \frac{d-2}{2} + \gamma(\lambda), \quad \gamma(0) = 0,$$

because canonical dimension of φ is $(d-2)/2$. Our nearest goal is to determine the RG functions $\beta(\lambda)$ and $\gamma(\lambda)$ for $d < 4$. The first one is particularly important because the RG flow along the massless part of $U(\mathcal{A}_{*G})$ (the trajectory C in the picture above) is described by the equation

$$\frac{d}{dl} \lambda = -\beta(\lambda).$$

We can calculate the correlation functions perturbatively, order by order in λ , and then find β and γ from the CS equation (21.4) itself, as we did before in the case $d = 4$. It is convenient to write the renormalized coupling constant as

$$\lambda = \mu^{4-d} \bar{\lambda},$$

where now $\bar{\lambda}$ is dimensionless parameter. Then by dimensional counting the N -point correlation functions have the form

$$\langle \varphi(x_1) \cdots \varphi(x_N) \rangle_\lambda = \mu^{N \frac{d-2}{2}} C^{(N)}(\mu x_1, \cdots, \mu x_N | \bar{\lambda})$$

and therefore the CS equation can be rewritten as

$$\left[\mu \frac{\partial}{\partial \mu} + N \bar{\gamma}(\bar{\lambda}) + \bar{\beta}(\bar{\lambda}) \frac{\partial}{\partial \bar{\lambda}} \right] \langle \varphi(x_1) \cdots \varphi(x_N) \rangle_{\bar{\lambda}} = 0, \quad (21.5)$$

where $\bar{\beta}$ and $\bar{\gamma}$ are the dimensionless functions of the dimensionless parameter $\bar{\lambda}$ defined as

$$\beta(\lambda) = \mu^\epsilon \bar{\beta}(\lambda/\mu^\epsilon), \quad \gamma(\lambda) = \bar{\gamma}(\lambda/\mu^\epsilon),$$

and the subscript $\bar{\lambda}$ indicates that the correlation function is expressed through $\bar{\lambda}$ instead of the dimensional λ . In what follows I will omit all these bars and use λ for $\bar{\lambda}$, β for $\bar{\beta}$, etc.; anyhow, these quantities coincide if μ is taken to be the unit of mass. The equation (21.5) can also be rewritten in terms of the momentum-space proper vertices,

$$\left[\mu \frac{\partial}{\partial \mu} - N \gamma(\lambda) + \beta(\lambda) \frac{\partial}{\partial \lambda} \right] \Gamma^{(N)}(p_1, \cdots, p_N) = 0. \quad (21.6)$$

We can now repeat our calculations of the leading perturbative contributions to the proper vertices, which of course are identical to the calculations within the dimensional regularization method, and obtain

$$\Gamma^{(2)}(p^2) = p^2 + O(\lambda^2),$$

and

$$\Gamma^{(4)}(p_1, \cdots, p_4) = \mu^{4-d} \left\{ \lambda - \frac{1}{2} \frac{\lambda^2}{(4\pi)^{\frac{d}{2}}} \frac{\Gamma(2-d/2)\Gamma^2(d/2-1)}{\Gamma(d-2)} \left[\left(\frac{p_{12}^2}{\mu^2} \right)^{\frac{d-4}{2}} + \left(\frac{p_{13}^2}{\mu^2} \right)^{\frac{d-4}{2}} + \left(\frac{p_{14}^2}{\mu^2} \right)^{\frac{d-4}{2}} - 3(2^{d-4}) \right] + O(\lambda^3) \right\}. \quad (21.7)$$

In the last equation the counterterm contribution is fixed by the normalization condition (21.3), i.e.

$$\Gamma^{(4)}(p_1, \cdots, p_4)|_{[(p_i+p_j)^2=4\mu^2]} = \mu^{4-d} \lambda.$$

Let me remind you that in dimensional regularization of $d = 4$ theory the counterterm contribution was introduced to cancel the $d = 4$ singularity of the Gamma-function $\Gamma(2 - d/2) \sim 2/(4 - d)$. For $d < 4$ the loop diagram contributions are finite even without any counterterm. We can expand in terms of the bare coupling constant λ_0 instead of the renormalized one; in this expansion the counterterm contribution in (21.7) would be absent. It is instructive to remember however that when $\epsilon = 4 - d$ is small the coefficients of this expansion in terms of the bare coupling constant are large,

$$\Gamma^{(4)}(p_1, \dots, p_4) \sim \lambda_0 - \frac{1}{\epsilon} \frac{\lambda_0^2}{(4\pi)^2} \left((p_{12}^2)^{-\epsilon} + (p_{13}^2)^{-\epsilon} + (p_{14}^2)^{-\epsilon} \right) + \dots \quad (21.8)$$

At $d = 4 - \epsilon < 4$ the "coordinate transformation" from λ_0 to λ is nonsingular in some neighborhood of 0, but at small ϵ it is "almost singular"²⁴.

Substituting (21.7) into (21.6) we find $\gamma(\lambda) = O(\lambda^2)$ and

$$\beta(\lambda) = (d - 4) \lambda + 3 \frac{\lambda^2}{(4\pi)^{\frac{d}{2}}} \frac{\Gamma(3 - d/2) \Gamma^2(d/2 - 1)}{\Gamma(d - 2)} + O(\lambda^3).$$

The term $(d - 4) \lambda$ appeared because of the factor μ^{4-d} ; it represents the canonical dimension of the coupling constant λ in the action (21.2).

Let us assume now that $d = 4 - \epsilon$ where ϵ is small,

$$\epsilon \ll 1.$$

Then

$$\beta(\lambda) = -\epsilon \lambda + 3 \frac{\lambda^2}{(4\pi)^2} + O(\lambda^3).$$

The linear term $-\epsilon \lambda$ brings in an important difference from the case $d = 4$. For small ϵ the RG flow equation

$$\frac{d}{dl} \lambda = -\beta(\lambda)$$

has nontrivial fixed point

$$\lambda_* = \frac{(4\pi)^2}{3} \epsilon.$$

The RG flow looks like this

²⁴The radius of convergence of the series $\lambda_0(\lambda)$ is $\sim \epsilon$.



This fixed point was obtained in the leading order of the renormalized perturbation theory. Note however that for small ϵ the fixed point value λ_* is small. The higher order terms can shift its position by terms $\sim \epsilon^2$,

$$\lambda_* = \frac{(4\pi)^2}{3} \epsilon + O(\epsilon^2), \quad (21.9)$$

but for sufficiently small ϵ these corrections cannot destroy the fixed point. The fixed point (21.9) is the Wilson-Fisher fixed point.

The above result was obtained using renormalized perturbation theory for $d < 4$ φ^4 theory. As I mentioned, for $d < 4$ the unrenormalized perturbation theory, i.e. the expansion in terms of the bare coupling constant λ_0 , is (essentially) finite. It is instructive to use this example to explain the relation between renormalized and unrenormalized perturbation theories.

If we repeat our calculation of the β -function using unrenormalized perturbation theory, i.e. with the expression (21.8) for the four-point vertex, we would obtain from (21.6)

$$\beta_0(\lambda_0) = -\epsilon \lambda_0, \quad (21.10)$$

with no fixed point visible. Higher order corrections do not change this result; in fact, it is easy to see that in unrenormalized perturbation theory, without the counterterm contributions, the formula (21.10) is exact. If one uses λ_0 as the coordinate on $U(\mathcal{A}_{*G})$, the RG function β reduces to its linear part. The bare coupling constant λ_0 is the “canonical coordinate”, whose existence is guaranteed by the Poincaré-Dulac theorem.

Indeed, in the case $\epsilon \ll 1$ the RG flow equation for small values of the renormalized parameter λ has the form

$$\frac{(4\pi)^2}{3} \frac{d}{dl} \lambda = \lambda (\lambda_* - \lambda). \quad (21.11)$$

It is not difficult to check that the change of variables

$$\lambda = \frac{\lambda_0}{1 + \frac{\lambda_0}{\lambda_*}} \quad (21.12)$$

linearizes this flow equation,

$$\frac{d}{dl} \lambda_0 = \epsilon \lambda_0. \quad (21.13)$$

Note that the transformation (21.12), or equivalently

$$\frac{1}{\lambda} = \frac{1}{\lambda_0} + \frac{1}{\lambda_*},$$

maps the interval $0 < \lambda < \lambda_*$ to the whole positive half-axis of λ_0 ,

$$\lambda \in [0, \lambda_*) \quad \leftrightarrow \quad \lambda_0 \in [0, \infty),$$

with the fixed point λ_* mapped to the point $\lambda_0 = \infty$; for this reason it is invisible in (21.13). Note also that if one expands λ in terms of λ_0 according to (21.12), for small ϵ the coefficients of this expansion become large,

$$\lambda = \lambda_0 - \frac{\lambda_0^2}{\lambda_*} + \dots,$$

which explains the large coefficients in (21.8). For $\epsilon = 0$ (i.e. $d = 4$) the interval $[0, \lambda_*)$ shrinks to zero, the perturbation becomes marginal, and the “canonical coordinate” λ_0 ceases to exist.

If the renormalized parameter λ is used, for small ϵ the flow between $\mathcal{A}_{*G}$ and $\mathcal{A}_{*WF}$, as well as the fixed point $\mathcal{A}_{*WF}$ itself, remains at small $\lambda \sim \epsilon$, within the domain where the perturbative results are reliable. Therefore for small ϵ we can use the renormalized perturbation theory to calculate all characteristics of the Wilson-Fisher fixed point. For instance, to the leading order in ϵ the flow equation for the mass parameter M^2 has the same form as at $d = 4$,

$$\frac{d}{dl} M^2 = \left(2 - \frac{\lambda}{(4\pi)^2} + O(\epsilon^2) \right) M^2. \quad (21.14)$$

Together with (21.11) this equation yields the pattern of the flow shown in the picture above.

ϵ -expansion

The main assumption of ϵ -expansion is that the general features of the RG flow observed at small ϵ remain qualitatively the same at $\epsilon = 1$, i.e. at $d = 4$. Under this

assumption, the characteristics of the Wilson-Fisher fixed point can be calculated perturbatively, as the power series in ϵ . Although at $d = 3$ the expansion parameter ϵ is not small, this approach yields remarkably accurate results for the critical exponents in three dimensions.

To find the critical exponents one must calculate the dimensions of various relevant fields at the Wilson-Fisher fixed point. Note that $\Phi_0 = \varphi^2$ is the only symmetric relevant field at $\mathcal{A}_{*WF}$. The dimensions of the fields Φ_0 and φ at this point can be calculated as

$$D_0 = D_{\varphi^2} = d - 2 + \gamma_{\varphi^2}(\lambda_*) ,$$

$$D_\varphi = \frac{d-2}{2} + \gamma(\lambda_*) .$$

In the leading order in ϵ we can just borrow the functions γ from the $d = 4$ calculation,

$$\gamma_{\varphi^2}(\lambda) = \frac{\lambda}{(4\pi)^2} + O(\lambda^2) ,$$

$$\gamma(\lambda) = \frac{1}{12} \frac{\lambda^2}{(4\pi)^4} + O(\lambda^3) ,$$

and use the value (19.7) for λ_* . This yields for the critical exponents

$$\frac{1}{\nu} = d - D_0 = 2 - \frac{\epsilon}{3} + O(\epsilon^2) ,$$

$$\alpha = 2 - \frac{d}{d - D_0} = \frac{\epsilon}{6} + O(\epsilon^2) ,$$

and similar results for other exponents. For $\epsilon = 1$ this gives $\nu = 0.60$ and $\alpha = 0.16$ for the correlation length and heat capacity exponents, to be compared to $\nu = 0.63$ and $\alpha = 0.12$ found in experiment ²⁵.

$O(N)$ φ^4 theory

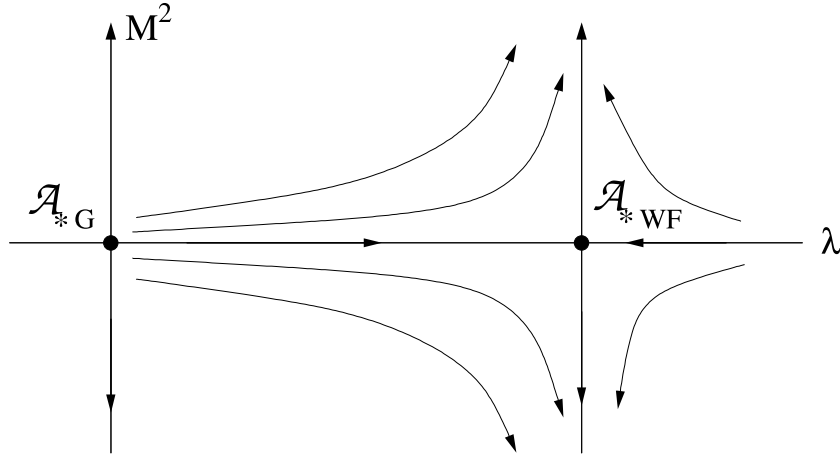
Let us discuss somewhat more general φ^4 theory. Assume that the scalar field has N components $\vec{\varphi} = (\varphi_1, \dots, \varphi_N)$, and the φ^4 interaction respects the $O(N)$ transformations of the vector $\vec{\varphi}$. We have

$$\mathcal{A} = \int d^d x \left[\frac{1}{2} (\partial \vec{\varphi})^2 + \frac{M^2}{2} \vec{\varphi}^2 + \frac{\lambda}{4!} (\vec{\varphi}^2)^2 \right]$$

²⁵The definition is $R_c \sim |T_c - T|^{-\nu}$, $C \sim |T_c - T|^{-\alpha}$.

Although, as before, we use the notation $M^2/2$ for the coefficient in front of the term $\vec{\varphi}^2$, this coefficient is allowed to take negative values.

At small $d = 4 - \epsilon$ with small ϵ the pattern of the RG flow of the $O(N)$ φ^4 theory in the vicinity of its Gaussian fixed point is essentially the same as in the one-component case we discussed previously²⁶.



There is the $O(N)$ version of the Wilson-Fisher fixed point $\mathcal{A}_{*,WF} \in \Sigma_c$ at $M^2 = 0$, $\lambda = \lambda_* \sim \epsilon$. In the space of $O(N)$ invariant interactions, it has one-dimensional unstable manifold $U(\mathcal{A}_{*,WF})$.

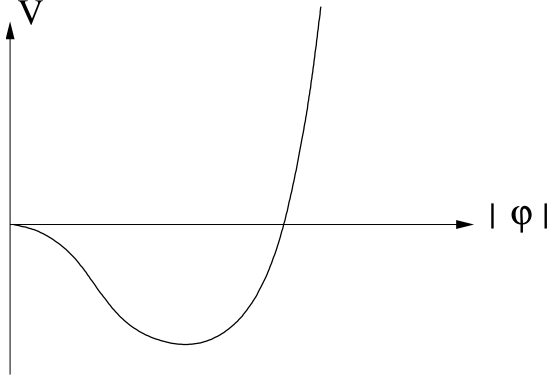
In the above picture the axis $M^2 = 0$ is the critical surface Σ_c . Above it, the flow generally goes upward, while everything below it flows downward. One can try to figure out the ultimate fate of the flow at positive and negative M^2 on physical grounds. The domain $M^2 > 0$ corresponds to the high-temperature phase of the ferromagnetic system. The pattern of the flow shows that M^2 increases with l . At large positive M^2 the term $\frac{M^2}{2} \vec{\varphi}^2$ suppresses fluctuations of $\vec{\varphi}$. Therefore at $M^2 > 0$ one expects to have attractive non-critical fixed point P_{high} , corresponding the probability distribution

$$P_{\text{high}}[\vec{\varphi}] \sim \prod_x \delta(\vec{\varphi}(x)).$$

²⁶Explicit calculation (Problem 9) yields

$$\beta(\lambda) = -\epsilon \lambda + \frac{N+8}{48\pi^2} \lambda^2 + \dots, \quad \gamma_{\vec{\varphi}^2}(\lambda) = \frac{N+2}{48\pi^2} \lambda + \dots$$

The domain of negative M^2 is associated with the low-temperature phase. Here under the RG flow M^2 remains negative while increasing in the absolute value. In this domain the potential term $\frac{M^2}{2} \vec{\varphi}^2 + \frac{\lambda}{4!} (\vec{\varphi}^2)^2$, as the function of $\vec{\varphi}^2$, has the minimum away from zero,



In terms of $\vec{\varphi}$ we have manifold of degenerate minima, the sphere $S^{(N-1)}$, of the radius

$$|\vec{\varphi}| = \rho := \sqrt{\frac{6|M^2|}{\lambda}}.$$

Points of the sphere indicate possible directions of $\vec{\varphi}$ at the minimum. It is easy to see that if $|M^2|$ is large the minimum in the above Figure becomes deep and narrow. Fluctuations of the radial part, $|\vec{\varphi}|$, are then suppressed. On the other hand there is no suppression for the fluctuations of the angular part. Writing

$$\vec{\varphi} = \rho \vec{n}, \quad |\vec{n}| = 1.$$

If we completely ignore the radial fluctuations, i.e. set $|\vec{\varphi}| = \rho$ everywhere, the action takes simple form

$$\mathcal{A} = \int d^d x \left[\frac{1}{2g} (\partial \vec{n})^2 \right]$$

with

$$g = \frac{1}{\rho} = \frac{\lambda}{6|M^2|}.$$

As $|M^2|$ increases with the RG flow, g becomes small ²⁷. Due to the non-linear

²⁷This simple classical analysis is valid only in the limit $\lambda \rightarrow 0$. More generally, fluctuations of $|\vec{\varphi}|$ can be taken into account, along with the fluctuations of the angular part, by renormalized perturbation theory. Some details about this theory - the so called linear sigma model - can be found in Chapter 11 of PS.

constraint $\vec{n}^2 = 1$ the above action leads to a theory more complicated than a free massless field. It is named the non-linear sigma model.

Let us analyze physical content of the nonlinear sigma-model. Classically, the most favorable field configurations correspond to minima of the action, which in this case are constant (x -independent) fields

$$\vec{n}(x) = \vec{n}_0.$$

By the $O(N)$ symmetry we can choose

$$\vec{n}_0 = (0, 0, \dots, 0, 1)$$

with no loss of generality. To take into account fluctuations around this minimum one can set

$$\vec{n}(x) = (\pi^1(x), \pi^2(x), \dots, \pi^{N-1}(x), \sigma(x)) \equiv (\pi^k(x), \sigma(x)) \quad (k = 1, 2, \dots, N-1)$$

with

$$\sigma(x) = \sqrt{1 - (\pi^k(x))^2}$$

The action takes the form

$$\mathcal{A} = \int d^d x \frac{1}{2g} \left[(\partial_\mu \pi^k)^2 + \frac{(\pi^k \partial_\mu \pi^k)^2}{1 - (\pi^k)^2} \right]$$

Ignoring first the quartic and higher terms in this action, we observe $N-1$ massless scalar fields $\pi^k(x)$. These are the *Goldstone bosons*, associated with the spontaneous breakdown of the $O(N)$ symmetry, down to $O(N-1)$ (Generalities about the spontaneous symmetry breaking are in Chapter 11 of PS).

Renormalizing

$$\pi^k = \sqrt{g} \phi^k$$

one can write the above action as

$$\mathcal{A} = \int d^d x \left[\frac{1}{2} (\partial \phi^k)^2 + \sum_{n=1}^{\infty} g^n O_n(x) \right],$$

where

$$O_n = \frac{1}{2} (\phi^k \partial_\mu \phi^k)^2 (\phi^k \phi^k)^n.$$

If $d > 2$, it is correct to think of it as $N-1$ -component of massless free field (the first term), with some composite fields added. The dimensions of the fields O_n are

$$O_n : \quad D_n = 2 + (n+2)(d-2) = d + (n+1)(d-2).$$

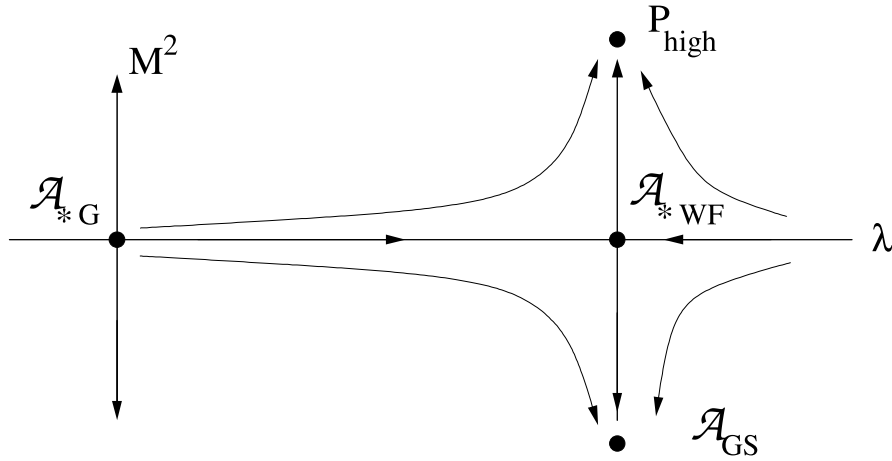
At $d > 2$ all these fields are irrelevant.

By these arguments, at $d > 2$ the long-distance physics in the low-temperature domain $M^2 < 0$ is controlled by the action

$$\mathcal{A}_{*,GS} = \int \frac{1}{2} (\partial\phi^k)^2 d^d x$$

which is essentially a Gaussian fixed point of $N - 1$ massless bosons. I will call it the "Goldstone point", to distinguish from the original Gaussian fixed point $\mathcal{A}_{*,G}$ (which has N massless scalar fields).

Returning to the pattern of the RG flow on $U(\mathcal{A}_{*,G})$, we now anticipate the following global structure



For $N > 1$ there is the *critical* fixed point $\mathcal{A}_{*,GS}$, which attracts the trajectories in the low-temperature domain $M^2 < 0$.

At $d = 2$ we have $D_n = 2 = d$, that is all the above fields O_n become marginal. Obviously, something interesting is happening in this case. Let us analyze it separately.

22 Nonlinear Sigma Model at $d = 2 + \epsilon$

Now I would like to discuss how the RG transformation acts on the action of the σ -model at $d = 2$. In this case the coupling constant g in

$$\mathcal{A} = \int d^2x \frac{1}{2g} (\partial \mathbf{n})^2$$

is dimensionless. I assume that g is small, so that the fluctuations of $\mathbf{n}(x)$ are small, since at small g the exponential factor $\exp(-\mathcal{A})$ strongly suppresses configurations having large gradients $\partial \mathbf{n}$. I remind that the field $\mathbf{n}(x)$ satisfies the non-linear constraint $\mathbf{n}^2(x) = 1$.

In principle, it is not difficult to develop renormalized perturbation theory in the coupling constant g , derive the CS equation and find the series expansion of associated flow function $\beta(g)$. This calculation is displayed in § 13.3 of PS. I will discuss the derivation based explicitly on the idea of integrating out the fast modes (also described in § 13.3 of PS), because it has some novel features as compared to what we have done before.

First, we introduce cutoff,

$$\mathbf{n}(x) = \int_{|k| < \Lambda} \mathbf{n}_k e^{ikx} \frac{d^2k}{(2\pi)^2}.$$

Note that now the modes $\mathbf{n}_k$ are not all independent, but must obey complicated constraint $\int_q \mathbf{n}_q \mathbf{n}_{k-q} = \delta^{(2)}(k)$, because of the σ -model constraint $\mathbf{n}^2(x) = 1$.

Next, we want to split $\mathbf{n}(x)$ into its “slow” and “fast” components. The above nonlinear constraint makes this task less straightforward than we had it before in usual scalar theories. Physically transparent way to make the splitting is as follows. We define the “slow” component $\mathbf{n}_0(x)$ as the field of unit length,

$$\mathbf{n}_0^2(x) = 1, \tag{22.1}$$

whose Fourier transform is restricted to $|k| < \Lambda/L$. To make further arguments transparent, I will assume that $L \gg 1$; we will see that in this case the dominating contribution to the RG transformed action comes from the integration over the fast modes with $\Lambda/L \ll |k| \ll \Lambda$. More precisely, I will assume that $L \gg 1$, but g is so small that

$$g \log L \ll 1.$$

At each spatial point x , one can choose a system of vectors $\mathbf{e}_i(x)$, $i = 1, 2, \dots, N-1$, such that N vectors $(\mathbf{n}_0(x), \mathbf{e}_i(x))$ form a basis. It is convenient to take orthonormal basis,

$$(\mathbf{n}_0(x) \mathbf{e}_i(x)) = 0, \quad (\mathbf{e}_i(x) \mathbf{e}_k(x)) = \delta_{ik}. \tag{22.2}$$

Since $\mathbf{e}_i(x)$ only need to be orthogonal to the “slow” field $\mathbf{n}_0(x)$, one can always choose them to be “slow” fields as well, containing only fourier modes with $|k|$ of the order of Λ/L or smaller. The full field $\mathbf{n}(x)$ can be written as

$$\mathbf{n}(x) = \mathbf{n}_0(x) \sqrt{1 - (\phi_i)^2(x)} + \mathbf{e}^i(x) \phi_i(x). \quad (22.3)$$

This expression automatically satisfies the constraint $\mathbf{n}^2(x) = 1$. $N - 1$ fields $\phi_i(x)$ represent the fast modes; they contain fourier components with $|k|$ between Λ/L and Λ . It is important that when g is small we can assume $\phi_i(x)$ are small.

In order to compute the action, we need spatial derivatives of $\mathbf{n}(x)$. These involve the derivatives of the vectors $\mathbf{n}_0(x)$ and $\mathbf{e}_i(x)$, as well as of the fields ϕ_i . One can expand them in the basic vectors as

$$\partial_\mu \mathbf{n}_0(x) = B_\mu^i(x) \mathbf{e}_i(x), \quad (22.4)$$

$$\partial_\mu \mathbf{e}_i(x) = A_\mu^{ik}(x) \mathbf{e}_k(x) - B_\mu^i(x) \mathbf{n}_0(x), \quad (22.5)$$

where

$$B_\mu^i(x) = (\mathbf{e}_i \partial_\mu \mathbf{n}_0) = -(\mathbf{n}_0 \partial_\mu \mathbf{e}_i), \quad (22.6)$$

$$A_\mu^{ik}(x) = (\mathbf{e}_k \partial_\mu \mathbf{e}_i) = -(\mathbf{e}_i \partial_\mu \mathbf{e}_k). \quad (22.7)$$

Note that the fields A_μ^{ik} and B_μ^i are “slow”. In terms of these variables the action is

$$\mathcal{A} = \frac{1}{2g} \int d^2x \left[\left(\partial_\mu \sqrt{1 - \phi^2} - B_\mu^i \partial_\mu \phi_i \right)^2 + \left(\partial_\mu \phi_i - A_\mu^{ik} \phi_k + B_\mu^i \sqrt{1 - \phi^2} \right)^2 \right].$$

For small g one can neglect all terms beyond the terms quadratic in ϕ ,

$$\mathcal{A} = \mathcal{A}[\mathbf{n}_0] + \tilde{\mathcal{A}}[\mathbf{n}_0, \phi_i],$$

where the first term comes from

$$\frac{1}{2g} \int d^2x B_\mu^i B_\mu^i = \frac{1}{2g} \int d^2x (\partial_\mu \mathbf{n}_0)^2 = \mathcal{A}[\mathbf{n}_0],$$

and

$$\tilde{\mathcal{A}} = \frac{1}{2g} \int d^2x \left[\left(\partial_\mu \phi_i - A_\mu^{ik} \phi_k \right)^2 + B_\mu^i B_\mu^k \left(\phi_i \phi_k - \delta_{ik} \phi^2 \right) \right].$$

Our goal now is to integrate out the fast modes ϕ . We treat the term $(1/2g) \int (\partial\phi)^2$ as the unperturbed action, yielding the propagator for the ϕ -field

$$i \text{ --- } k = \frac{g \delta_{ik}}{p^2}.$$

The remaining terms are treated as perturbations, giving rise to vertices of three types:

$$\begin{array}{c}
 \begin{array}{c} j \\ \nearrow \\ \text{---} \bullet \text{---} A \\ \nwarrow \\ k \end{array}
 \end{array}
 \begin{array}{c}
 p_1 \\ \nwarrow \\ p_2
 \end{array}
 = \frac{i A_\mu^{jk}}{g} (p_1 - p_2)_\mu,$$

$$\begin{array}{c}
 \begin{array}{c} j \\ \nearrow \\ \text{---} \bullet \text{---} A^2 \\ \nwarrow \\ k \end{array}
 \end{array}
 = - \frac{A_\mu^{ij} A_\mu^{ik}}{g},$$

$$\begin{array}{c}
 \begin{array}{c} j \\ \nearrow \\ \text{---} \bullet \text{---} B^2 \\ \nwarrow \\ k \end{array}
 \end{array}
 = \frac{\delta_{jk} B_\mu^i B_\mu^i - B_\mu^j B_\mu^k}{g}.$$

As usual

$$\mathcal{A}_1 = \mathcal{A} - \text{sum of all connected diagrams},$$

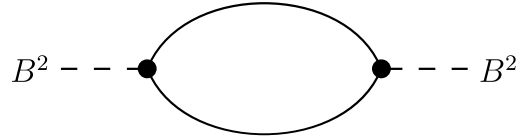
with the dotted lines taken as the external legs.

The leading order in g comes from the one-loop diagrams. The simplest (and the most important) of those is

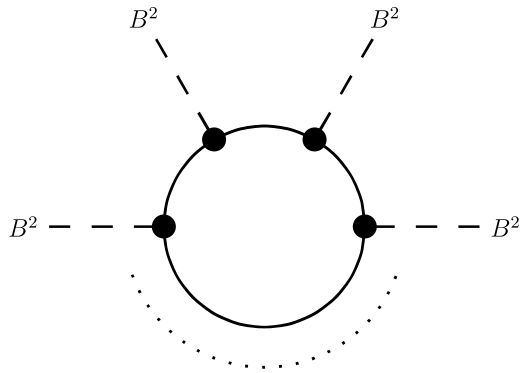
$$\begin{aligned}
 B^2 \text{ --- } \text{---} \bullet \bigcirc &= \frac{\delta_{jk} B_\mu^i B_\mu^i - B_\mu^j B_\mu^k}{2g} \int \frac{d^2 p}{(2\pi)^2} \frac{g \delta_{jk}}{p^2} = \\
 &= \frac{(N-2)(B_\mu^i)^2}{2} \frac{1}{4\pi} \int_{\Lambda^2/L^2}^{\Lambda^2} \frac{dp^2}{p^2} = \frac{N-2}{4\pi} (\partial_\mu \mathbf{n}_0)^2 \log L.
 \end{aligned}$$

Thus, this diagram brings in a term identical in the form to the original action, i.e. it renormalizes the coupling parameter g .

The diagram

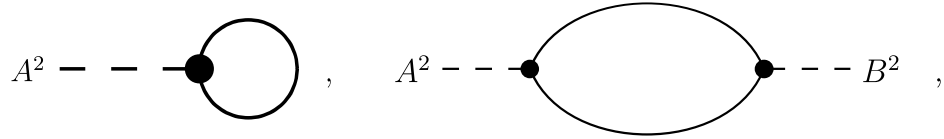


generates term $\sim B^4$ (with the coefficient $\sim 1/\Lambda^2$); the field B^4 has dimension $4 > d$ and hence it is an irrelevant field; it can be neglected in the leading approximation. Likewise, all the diagrams of the form



can be ignored, for the same reason.

Let us take a look at the diagrams with A -vertices. Consider the following ones



Both these diagrams show logarithmic behavior,

$$\int \frac{d^2 k}{k^2} \sim \log L, \quad \text{and} \quad \int \frac{d^2 k}{k^2} \frac{k^\mu k^\nu}{k^2} \sim \log L$$

However, it is not difficult to show that these two logarithmic contributions in fact cancel each other. One can do it by a direct calculation, but it is easy to prove this statement on the basis of certain symmetry.

The $\log L$ of each of these two diagrams is of the form

$$\log(L) A_\mu^{ij} A_\mu^{ij},$$

where, as before

$$A_\mu^{ij} = \mathbf{e}^j \partial_\mu \mathbf{e}^i$$

Recall that $\mathbf{e}^i(x)$ are arbitrary vectors tangent to $S^{(N-1)}$ at $\mathbf{n}_0(x)$. They satisfy

$$\mathbf{e}^i \mathbf{n}_0 = 0, \quad \mathbf{e}^i \mathbf{e}^j = \delta^{ij},$$

but otherwise arbitrary. One can always choose them differently,

$$\tilde{\mathbf{e}}^i(x) = \Omega_{ij}(x) \mathbf{e}^j(x),$$

where $\Omega_{ij}(x)$ is orthogonal $(N-1) \times (N-1)$ matrix,

$$\Omega(x) \Omega^t(x) = I.$$

$\Omega(x)$ is "slow" (for $\mathbf{e}^i(x)$ were assumed to be slow), but otherwise arbitrary function of x . Under this local orthogonal transformations, $A_\mu^{ij}(x)$ as a $O(N-1)$ gauge potential,

$$\tilde{A}_\mu(x) = \Omega(x) A_\mu(x) \Omega^t(x) + \Omega(x) \partial_\mu \Omega^t(x),$$

where $A_\mu(x)$ stands for $(N-1) \times (N-1)$ anti-symmetric matrix $A_\mu^{ij}(x)$.

As this gauge transformation describes just the freedom in the choice of variables, the action $\mathcal{A}_1[\mathbf{n}_0]$ must be gauge invariant. It is easy to check that the combination

$$A_\mu^{ij} A_\mu^{ij} = \text{tr}(A_\mu A_\mu^t)$$

is not gauge invariant, and thus it can not appear in $\mathcal{A}_1$. The simplest gauge invariant local combination of course is

$$\text{tr}(F_{\mu\nu} F_{\mu\nu}),$$

with

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu].$$

However it involves extra derivative, and thus corresponds to an irrelevant term.

Therefore, in this approximation the only interesting contribution comes from the above loop diagram with single B^2 vertex. We have

$$\mathcal{A}_1 = \int \left(\frac{1}{2g} - \frac{N-2}{4\pi} \log L \right) (\partial_\mu \mathbf{n}_0)^2 d^2 x. \quad (22.8)$$

Thus, the step I of the RG transformation leads to

$$\frac{1}{g} \rightarrow \frac{1}{g_1} = \frac{1}{g} - \frac{N-2}{2\pi} \log L.$$

The step 2 is nearly trivial. If one changes

$$\mathbf{n}_0(x) = \mathbf{n}(x/L)$$

the action does not change. Therefore

$$RG_l : \quad \frac{1}{g} \rightarrow \frac{1}{g} - \frac{N-2}{2\pi} l.$$

In other words, the beta-function is

$$\beta(g) = -\frac{N-2}{2\pi} g^2 + O(g^3),$$

and for small g the RG flow equation

$$\frac{dg}{dl} = \frac{N-2}{2\pi} g^2$$

shows that $g(l)$ *increases* towards large distances; at $d = 2$ the Goldstone fixed point becomes unstable.

The above flow equation determines (in the leading order in g) the behavior of the action under the RG transformation. One might be interested in the behavior of correlation functions of, say, the field $\mathbf{n}(x)$. To that end one needs then to find the RG function $\gamma_{\mathbf{n}}(g)$ in the CS equation

$$\left(x \frac{\partial}{\partial x} + n \gamma_{\mathbf{n}}(g) + \beta(g) \frac{\partial}{\partial g} \right) \langle \mathbf{n}(x_1) \dots \mathbf{n}(x_n) \rangle_c = 0.$$

It is defined in terms of the action of the RG transformation on the field $\mathbf{n}(x)$.

Consider any correlation function involving $\mathbf{n}(x)$,

$$\langle \mathbf{n}(x) \dots \rangle,$$

and assume again that the momentum p associated with x is $\ll \Lambda/L$. Then, splitting as before

$$\mathbf{n} = \mathbf{n}_0 \sqrt{1 - \phi^2} + \mathbf{e}^i \phi_i \simeq \mathbf{n}_0 \left(1 - \frac{1}{2} \phi^2 \right) + \mathbf{e}^i \phi_i$$

one needs to find expectation value of this combination over the ensemble of the fast modes ϕ . The last term can be ignored (its expectation value is zero), and we have

$$\mathbf{n} \rightarrow \mathbf{n}_0 \left(1 - \frac{1}{2} \langle \phi_i^2 \rangle \right) := Z_{\mathbf{n}}^{-\frac{1}{2}}(L) \mathbf{n}_0,$$

where

$$Z_{\mathbf{n}}^{-\frac{1}{2}}(L) = 1 - \frac{1}{2} \langle \phi_i^2 \rangle = 1 - \frac{N-1}{2} \frac{g}{2\pi} l.$$

Since at $d = 2$ $\mathbf{n}$ has zero canonical dimension, we then have

$$\gamma_{\mathbf{n}}(g) = D_{\mathbf{n}}(g) = - \frac{d}{dl} Z_{\mathbf{n}}^{-\frac{1}{2}}(l) \Big|_{l=0} = \frac{N-1}{4\pi} g.$$

In the above calculation we have assumed that $gl \ll 1$; that is why the higher-order terms in ϕ were neglected. But, as usual, we can extend it to larger l . Generally, we have

$$RG_l \mathbf{n} = Z_{\mathbf{n}}^{-\frac{1}{2}}(l) \mathbf{n},$$

where $Z_{\mathbf{n}}^{-\frac{1}{2}}(l)$ is a solution of the equation

$$- \frac{d}{dl} Z_{\mathbf{n}}^{-\frac{1}{2}}(l) = \gamma_{\mathbf{n}}(g(l)) Z_{\mathbf{n}}^{-\frac{1}{2}}(l),$$

and $g(l)$ is the integral curve of the equation

$$\frac{dg(l)}{dl} = -\beta(g(l)) .$$

In our case

$$\frac{1}{g(l)} = \frac{1}{g(0)} - \frac{N-2}{2\pi} l .$$

Recall that the "flow" of $Z_{\mathbf{n}}^{-\frac{1}{2}}(l)$ determines the scale dependence of the correlation functions

$$\langle \mathbf{n}(x_1/L) \dots \mathbf{n}(x_n/L) \rangle_{g(l)} = Z_{\mathbf{n}}^{-\frac{n}{2}}(l) \langle \mathbf{n}(x_1) \dots \mathbf{n}(x_n) \rangle_{g(0)} ,$$

where $l = \log L$.

Exercise: For the $O(N)$ nonlinear sigma model,

I. At $d = 2$

a) Assuming $g(l) \ll 1$, find $Z_{\mathbf{n}}(l)$.

b) Derive the following $|x| \rightarrow 0$ asymptotic of the two-point correlation function

$$\langle \mathbf{n}(x) \mathbf{n}(0) \rangle \sim \left[\log \left(\frac{1}{x^2} \right) \right]^{\frac{N-1}{N-2}} .$$

Explain how this result is in agreement with the exact solution of the model at $N \rightarrow \infty$ (PS, §13.3, pages 463-465)

II. Find the flow equation for $g(l)$ and $Z_{\mathbf{n}}(l)$ at $d = 2 + \epsilon$, with $\epsilon \ll 1$. Again, determine the $|x| \rightarrow 0$ behavior of the two-point correlation function.

Coming back to the flow equation ($d = 2$)

$$\frac{dg}{dl} = \frac{N-2}{2\pi} g^2 + \dots$$

one observes that (at $N > 2$) the flow of the coupling $g(l)$ looks like this:



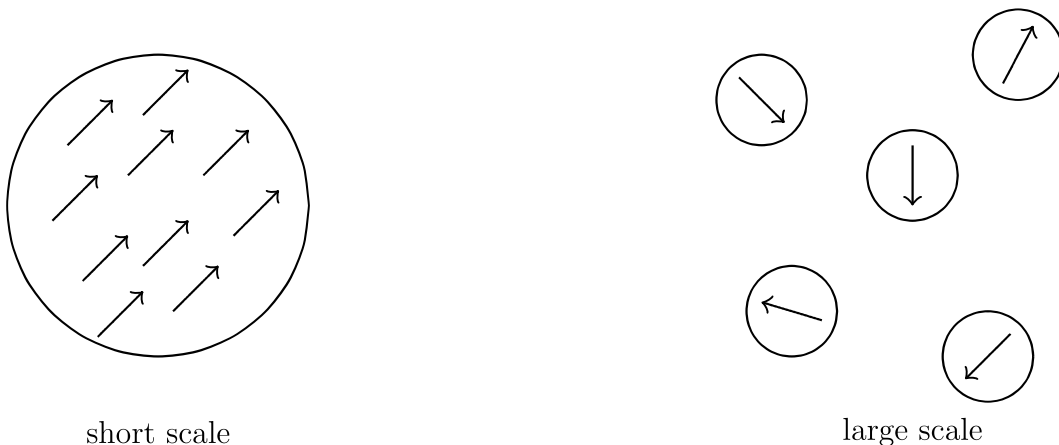
At large distances g increases, and the interaction becomes strong. Our weak coupling analysis is no longer valid (in particular the higher order terms in $\beta(g)$ can

not be neglected). On the other hand, if we follow the flow of $g(l)$ backward, to $l \rightarrow -\infty$,

$$g(l) = \left(\text{Const} - \frac{N-2}{2\pi} l \right)^{-1} \rightarrow \frac{2\pi}{N-2} (-l)^{-1},$$

the interaction of the Goldstone modes become weaker, and the short-distance behavior is controlled by a free theory. For this reason, this type of RG flow is called *asymptotic freedom*. Note that the situation is opposite to that in φ^2 theory at $d = 4$, where instead $\lambda \rightarrow 0$ at long scales, whereas perturbation theory breaks down at short distances.

The asymptotic freedom allows one to establish existence of local field theory; indeed it shows that at short scales the fluctuations are suppressed, and the cutoff can be removed with no violation of locality. On the other hand, at longer scales the fluctuations become stronger. It is not possible to describe these large scale fluctuations by perturbation theory. Nonetheless, one can make intuitive picture based on this pattern of the RG flow. If one concentrate attention on very short scales, the vectors $\mathbf{n}(x)$ are nearly parallel, because at $l \rightarrow -\infty$ $g(l)$ is small, thus suppressing the gradients of $\mathbf{n}(x)$. However, if one looks at larger scales, the directions of $\mathbf{n}(x)$ at the points separated by distance $\sim R_c$ are much less correlated, and if the separation is even much larger, there is no correlation.



This picture suggests that at $d = 2$ the phase with spontaneously broken $O(N)$ symmetry does not exist. In other words, $d = 2$ magnet has no ferromagnetic phase with spontaneous magnetization. Let me stress that the perturbative analysis of the RG flow does not prove this statement, which requires much more involved

arguments. Note also that this picture is expected to apply only if $N > 2$ ²⁸.

What is different at $d > 2$? This question can be studied in details in formal case of $d = 2 + \epsilon$ with $\epsilon \ll 1$. In this case we can repeat our calculation of the RG flow at small g . The step 1 (elimination of the fast part) yields essentially the same result,

$$\mathcal{A}_1 = \int \left(\frac{1}{2g} - \frac{N-2}{4\pi} \Lambda^\epsilon \log L \right) (\partial_\mu \mathbf{n}_0)^2 d^2x . \quad (22.9)$$

However, the step 2 (scale transformation) is no longer exactly trivial. The change of variables

$$\mathbf{n}_0(x) = \mathbf{n}(x/L)$$

leads to

$$\int (\partial_\mu \mathbf{n}_0)^2 d^d x = L^{d-2} \int (\partial_\mu \mathbf{n})^2 d^d x .$$

Therefore

$$\mathcal{A}' = \int L^\epsilon \left(\frac{1}{2g} - \frac{N-2}{4\pi} \Lambda^\epsilon \log L \right) (\partial_\mu \mathbf{n}_0)^2 d^2x .$$

Since

$$L^\epsilon \approx 1 + \epsilon \log L$$

one finds that under the RG transformation

$$\frac{1}{g} \rightarrow \frac{1}{g} + \left(\frac{\epsilon}{g} - \frac{N-2}{2\pi} \right) \log L ,$$

²⁸At $N = 2$ the g^2 term in the β -function vanishes. In fact, it is easy to see that all higher terms vanish as well. At $N = 2$ the direction $\mathbf{n}(x)$ can be parameterized by a single angle $\theta(x)$, $\mathbf{n}(x) = (\cos \theta(x), \sin \theta(x))$, and formally

$$\mathcal{A} = \frac{1}{2g} \int (\partial_\mu \theta)^2 d^2x ,$$

which is just the action of free massless theory. The action of RG on a free theory is trivial. However, beyond the perturbation theory, the $N = 2$ sigma model has more complex structure. This is because the field $\theta(x)$ is an angular variable, $\theta(x) \sim \theta(x) + 2\pi N(x)$, where $N(x)$ is integer-valued. As the result, there are topologically nontrivial field configurations ("vortices", or "instantons"). Their interaction leads to remarkable phase transition at finite $g = g_c$ - the so called Berezinsky - Kosterlitz - Thouless transition (See e.g. Kosterlitz, Thouless, "Ordering, metastability and phase transitions in two-dimensional systems", Journal of Physics C: Solid State Physics, Vol. 6 pages 1181-1203 (1973)).